Local Existence and Covariance for Slope-Prescribed Nearly-Vacuum Metrics

Corrected Proof Sketch for gPCT Reconstruction

Christopher Dean White
ORCID: 0009-0002-7866-7078

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Abstract

We present a corrected and refined analytic outline of the local existence and covariance lemma for gravitational phase-cancellation theory (gPCT). Given a smooth slope history $s(\tau)$ on a compact interval satisfying quantitative smallness and finite-Hilbert-transform centering conditions, we construct a Lorentzian metric in a tubular neighborhood of a worldline such that $\frac{1}{2}\ln(-g_{00})$ reproduces s along the worldline while the Einstein tensor remains $O(\epsilon)$. The construction employs a dual-projection decomposition in Gaussian normal coordinates, a Banach fixed-point argument via interior Schauder estimates, and Carleman-based unique continuation to establish chart-independent covariance. All projections, gauges, and operators are defined explicitly.

1 Statement

Let $I \subset \mathbb{R}$ be a compact interval of length $L \le L_0$ and let $s \in C^\infty(I)$ satisfy

$$\Big|\int_I s(\tau)\,d\tau\Big| \le \epsilon, \qquad \big\|\mathcal{H}_I(s) + \tfrac{1}{2\pi}\big\|_{L^\infty(I)} \le \epsilon \tag{1}$$

for a sufficiently small dimensionless parameter $\epsilon \in (0, \epsilon_0)$. Here $\mathcal{H}_I$ denotes the finite Hilbert transform on I,

$$\mathcal{H}_I(s)(\tau) = \frac{1}{\pi}\,\mathrm{P.V.}\int_I \frac{s(\sigma)}{\tau - \sigma}\,d\sigma. \tag{2}$$

(The principal-value integral is well-defined for $C^{1,\alpha}$ functions; the centering condition is the quantitative form of the White attractor.)

Then there exists a four-dimensional Lorentzian metric $g_{\mu\nu}$ of class $C^{2,\alpha}$ on a tubular neighborhood $\mathcal{U}_\delta = I \times B^3_\delta(0)$ of the base worldline $\Gamma = \{x^i = 0\}$ such that:

1. $\frac{1}{2}\ln(-g_{00})\big|_\Gamma = s(\tau)$;
2. $\|G_{\mu\nu}(g)\|_{L^\infty(\mathcal{U}_\delta)} \le C\epsilon$ for a structural constant C independent of absolute scales;
3. the bound is stable under admissible quasi-harmonic gauge transformations (covariance).

The metric is unique up to local diffeomorphisms of class $\mathrm{Diff}^{3,\alpha}$.

2 Setup and Dual Projection

Work in Gaussian normal coordinates (τ, x^i) adapted to Γ, so that

$$g = -e^{2\phi_g(\tau,x)}\,d\tau^2 + \gamma_{ij}(\tau,x)\,dx^i\,dx^j, \qquad g_{0i} = 0, \tag{3}$$

with boundary conditions

$$\phi_g(\tau,0)=s(\tau),\qquad \gamma_{ij}(\tau,0)=\delta_{ij}. \tag{4}$$

(The coordinate τ is proper time along Γ.)

Definition 1 (Dual projection)**.** *Let $\mathbf{h}=(\phi_g,\gamma_{ij})$ be a metric perturbation relative to the flat background. Define the longitudinal projection $\mathbf{P}_\parallel$ by restricting to the worldline normal direction (Fermi–Walker transport along Γ) and the transverse projection by*

$$\mathbf{P}_\perp=\mathrm{Id}-\mathbf{P}_\parallel. \tag{5}$$

Explicitly, on the spatial slices Σ_τ,

$$\mathbf{P}_\parallel\phi_g=\phi_g\big|_\Gamma,\qquad \mathbf{P}_\perp\gamma_{ij}=\gamma_{ij}-\delta_{ij}-\textit{(trace terms fixed by constraints).} \tag{6}$$

The dual projection enforces that transverse gradients of the lapse are controlled by the residual of the recursion scalar. Under the Hilbert-centering hypothesis the off-worldline residuals satisfy the a-priori bounds

$$\|\partial_i\phi_g\|_{L^\infty(\mathcal{U}_\delta)}\le C_1\epsilon,\qquad \|\partial_i\partial_j\phi_g\|_{L^\infty(\mathcal{U}_\delta)}\le C_2\epsilon \tag{7}$$

(to be closed by the fixed-point argument below).

3 Reduced Einstein System and Quadratic Remainder

The vacuum Einstein equations in Gaussian normal coordinates reduce to a quasilinear wave system for the spatial metric together with constraints. For the lapse one obtains the rough d'Alembertian equation

$$\Box_\gamma\phi_g=Q(\partial\phi_g,\partial^2\phi_g)+\mathcal{S}_{\mathrm{gauge}}, \tag{8}$$

where Q collects the quadratic Christoffel contractions

$$Q_{00}=\gamma^{ij}\partial_i\phi_g\partial_j\phi_g+\tfrac14 e^{-2\phi_g}(\gamma^{ik}\partial_\tau\gamma_{kj})(\gamma^{jl}\partial_\tau\gamma_{li})-\partial_\tau\phi_g\cdot\tfrac12\gamma^{ij}\partial_\tau\gamma_{ij}, \tag{9}$$

$$Q_{0i}=\partial_k\phi_g\cdot\tfrac12\gamma^{kl}\partial_\tau\gamma_{li}-\partial_i\phi_g\big(\partial_\tau\phi_g+\tfrac12\gamma^{kl}\partial_\tau\gamma_{kl}\big), \tag{10}$$

$$Q_{ij}=\partial_i\phi_g\partial_j\phi_g+{}^{(3)}R_{ij}(\gamma)-\tfrac12 e^{-2\phi_g}\partial_\tau^2\gamma_{ij}+O(|\nabla\phi_g|^2,|\partial_\tau\gamma|^2). \tag{11}$$

Because $\mathbf{P}_\perp\nabla\phi_g|_\Gamma=0$ by the boundary conditions, the cross-projection terms vanish on Γ. Interior Schauder estimates then yield

$$\|Q\|_{C^{0,\alpha}(\mathcal{U}_\delta)}\le K'\epsilon^2 \tag{12}$$

once the gradient bounds (7) are available.

The gauge source $\mathcal{S}_{\mathrm{gauge}}$ is fixed by the dual-projection constraint

$$\mathcal{F}(g)=\big(\mathbf{P}_\parallel(\Gamma^\mu{}_{\alpha\beta}g^{\alpha\beta}),\mathbf{P}_\perp(g_{0i})\big)=(0,0). \tag{13}$$

The differential $D\mathcal{F}$ at the Minkowski background is invertible on the gauge orbit, eliminating zero modes.

4 Existence by Banach Fixed Point

Define the closed ball in the Banach space

$$\mathcal{X}_\epsilon = \Big\{(\phi_g, \gamma_{ij}) \in C^{2,\alpha}(\mathcal{U}_\delta) : \phi_g\big|_\Gamma = s,\ \|\phi_g - s\|_{C^{2,\alpha}} \le M\epsilon,\ \|\gamma_{ij} - \delta_{ij}\|_{C^{2,\alpha}} \le M\epsilon\Big\}. \tag{14}$$

Let $\mathcal{G}_0$ be the Green's operator for the background wave operator with Dirichlet data on Γ. The solution map

$$\mathcal{T}(\phi, \gamma) = \mathcal{G}_0 \circ \big(Q(\partial\phi, \partial^2\phi) + \mathcal{R}_{\mathbf{P}}(\phi, \gamma)\big) \tag{15}$$

maps $\mathcal{X}_\epsilon$ into itself for $\epsilon \le \epsilon_0$ sufficiently small. Classical interior Schauder estimates give

$$\|\mathcal{T}(u)\|_{C^{2,\alpha}} \le C_S\big(\|Q(u)\|_{C^{0,\alpha}} + \|s\|_{C^{2,\alpha}}\big). \tag{16}$$

The quadratic nature of Q produces the contraction

$$\|\mathcal{T}(u) - \mathcal{T}(v)\|_{\mathcal{X}_\epsilon} \le \tfrac{1}{2}\|u - v\|_{\mathcal{X}_\epsilon}. \tag{17}$$

By the Banach fixed-point theorem there exists a unique fixed point $g \in \mathcal{X}_\epsilon$. The Einstein tensor of this metric satisfies

$$\|G_{\mu\nu}(g)\|_{L^\infty(\mathcal{U}_\delta)} \le C'\epsilon, \tag{18}$$

where the structural constant

$$C' \le C_S\big(1 + 2C_1^2\epsilon_0 + \tfrac{M}{2} + K_\gamma M^2\epsilon_0\big) \tag{19}$$

is independent of absolute mass or length scales (all quantities having been nondimensionalized by the interval length L).

5 Unique Continuation and Covariance

Let g_1, g_2 be two solutions realizing the same s. Their difference $\delta g = g_1 - \Phi^* g_2$ (after optimal gauge alignment) satisfies the linearized Einstein equation

$$P(\delta g) = \mathcal{E}(\delta g) = O(\epsilon). \tag{20}$$

Under the dual-projection splitting,

$$P = P_\parallel \oplus P_\perp + \mathcal{K}_{\text{cross}}, \qquad \|\mathcal{K}_{\text{cross}}\| \le C_{\mathcal{K}}\epsilon. \tag{21}$$

On the punctured cylinder introduce the strictly subharmonic weight $\varphi(x) = |x|^2$. For large λ the conjugated operator satisfies the Carleman estimate

$$\int_{\mathcal{U}_\delta} e^{2\lambda\varphi}|P(\delta g)|^2\, dV \ge C_0\lambda \int e^{2\lambda\varphi}|\nabla\delta g|^2\, dV + C_0\lambda^3 \int e^{2\lambda\varphi}|\delta g|^2\, dV. \tag{22}$$

Cross terms are absorbed by the bulk λ^3 term. Restricting to the concentric sub-ball $B_{r/2}$ yields

$$\|\delta g\|_{L^2(B_{r/2})} \le C\epsilon. \tag{23}$$

Elliptic bootstrapping on the spatial slices (modulo gauge) and Sobolev embedding give

$$\|\delta g\|_{C^{2,\alpha}(B_{r/2})} \le C_4\epsilon. \tag{24}$$

Thus $\psi^* g_2 - g_1 = O(\epsilon)$ for an explicit diffeomorphism $\psi \in \mathrm{Diff}^{3,\alpha}$, establishing chart-independent covariance. (The Carleman estimate is compatible with the known unique-continuation results for Einstein metrics in geodesic normal coordinates.)

6 Continuous Dependence and Gauge Anchoring

The gauge-fixing map $\mathcal{F}$ has invertible differential at the flat background, so

$$\ker(D\mathcal{F}(\eta)) \cap \ker(P) = \{0\}. \tag{25}$$

Consequently all coordinate freedom is anchored and spurious gauge modes are suppressed at rate $O(\epsilon)$. The solution map

$$\Xi : C^\infty(I) \to C^{2,\alpha}(\mathcal{U}_\delta), \qquad s \mapsto g_\epsilon \tag{26}$$

is continuous. In the limit $\epsilon \to 0$ one recovers the exact Minkowski vacuum:

$$\lim_{\epsilon \to 0} \|g_\epsilon - \eta\|_{C^{2,\alpha}(\mathcal{U}_\delta)} = 0. \tag{27}$$

7 Status

The dual-projection realization, finite-Hilbert-transform centering, and explicit gauge-fixing convert the quantitative estimates into a closed local-existence architecture. The principal remaining verification is the precise numerical range of the Schauder and Carleman constants for the concrete reduced operator; these follow standard lines in the local theory of the Einstein equation and do not affect the structural claims.